



Fig. 4: Muller pipeline

lowering the clock speed. But, switching the clock speed takes time and energy and rarely achieves optimal operation conditions. An asynchronous processor, in contrast, can switch from zero to 100 per cent operational speed immediately. In fact, there is no switching activity if the processor is idle.

Electromagnetic compatibility. In asynchronous processors, it is unlikely that the majority of transistors switch at the same point in time, which leads to much lower electromagnetic radiation than in synchronous processors because the pulsed electromagnetic radiation of millions of concurrently switching transistors of synchronous processors adds up to a large electromagnetic pulse. Still, asynchronous circuits tend to have a characteristic electromagnetic radiation

Reliability and robustness. If the environment of a processor gets rough, it has to have some kind of strategy to adapt itself to this situation and still keep working reliably and as fast as possible. For example, if the temperature of the processor increases, some element, e.g., full adder, may take a small but significantly longer time to compute the result than it would

take at lower temperatures. This must be taken into account to ensure reliability.

Synchronous processors do this by actively regulating their clock speed, whereas asynchronous processors don't have to do anything—it's done implicitly. An asynchronous processor uses delay lines and (possibly) wire-delay-insensitive handshaking protocols. If an element takes longer to compute its result, the delay line also takes longer to relay the 'request' signal to the successor element, as it is just as sensitive to the environment as the real element.

Drawbacks of asynchronous approach

The handshaking mechanism introduces overhead, which may slow down the circuit or even cost more energy than it saves. Also, there is lack of appropriate computer-aided design (CAD) tools. The commercially available tools are optimised for synchronous designs, which is a big disadvantage for designers.

As the pipeline stages do not work synchronously, it is difficult to exchange the state information between pipeline

stages, which leads to redundant data storage, i.e., the program counter has to be stored with every pipeline stage. This is not a significant overhead in energy consumption as most of the time only a few bits of the program counter change. Furthermore, the control logic for asynchronous processors is more complex than for synchronous processors, which leads to a comparatively high energy consumption of about 40 per cent in the case of AMULET3

core. This also includes most parts of the clock-counterpart.

To sum up

Asynchronous microprocessors have some specific advantages like very good electromagnetic compatibility and up to four times higher energy efficiency than functionally equivalent synchronous implementations, which makes them useful in niche areas like smartcards where low power and robustness are important. In the future, asynchronous processors may also benefit from techniques developed for synchronous processors, like lowering the supply voltage, which is easy to implement in asynchronous designs because of the flexibility that handshaking brings along.

In any case, more research and development is needed especially in the area of CAD tools where the commercially available applications are still sub-optimal. First steps are made by the Balsa language and CAD suite for asynchronous design. ●

The author is a research associate in Electrical Department, IIT Delhi, engaged in multimedia research

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